



Criteria for the diagnosis of cardiogenic shock—European perspective: A Clinical Consensus Statement of the Association for Acute CardioVascular Care, the Heart Failure Association, and the European Association of Percutaneous Cardiovascular Interventions of the ESC

Hoong Sern Lim ¹, Agnieszka Tycinska ², Clement Delmas^{3,4}, Jacob Eifer Moller⁵, Janine Pöss⁶, Christian Stoppe^{7,8,9,10}, Christophe Vandenbriele ^{11,12,13}, Holger Thiele ⁶, Kurt Huber ¹⁴, Ingo Ahrens^{15,16}, Christian Hassager ¹⁷, Ovidiu Chioncel ¹⁸, Marco Metra^{19,20}, Matteo Pagnesi ²¹, Wiktor Kuliczowski²², Maja Cikes ²³, Linda W. van Laake ²⁴, Alessandro Sionis ²⁵, Mario Iannaccone ²⁶, Stephane Manzo-Silberman²⁷, Federico Pappalardo ²⁸, Susanna Price ^{29,30}, Thomas F. Lüscher^{29,30,31,32}, and Benedikt Schrage ^{33,34,35*}

¹Department of Cardiology, University Hospitals Birmingham NHS Foundation Trust, Birmingham, UK; ²Department of Intensive Cardiac Care, Medical University of Białystok, Poland; ³Intensive Cardiac Care Unit, Cardiology Department, Toulouse University Hospital, Toulouse, France; ⁴REICATRA, Institut Saint Jacques, Toulouse, France; ⁵Department of Cardiology, Copenhagen University Hospital Rigshospitalet, Copenhagen, Denmark; ⁶Heart Center Leipzig at Leipzig University and Leipzig Heart Science, Leipzig, Germany; ⁷Department of Anaesthesiology, Intensive Care, Emergency and Pain Medicine, University Hospital Würzburg, Würzburg, Germany; ⁸Department of Cardiac Anesthesiology and Intensive Care Medicine, Deutsches Herzzentrum der Charité, Berlin, Germany; ⁹Charité—Universitätsmedizin Berlin, Corporate Member of Freie Universität Berlin and Humboldt-Universität zu Berlin, Berlin, Germany; ¹⁰Center for Outcomes Research and Department of Anesthesiology, UTHealth, Houston, TX, USA; ¹¹Heart Center Aalst, Azorg Hospital, Aalst, Belgium; ¹²Cardiac Intensive Care, Harefield Hospital, Guy's & St Thomas's NHS Foundation Trust, London, UK; ¹³Department of Surgery and Cancer, Imperial College, London, UK; ¹⁴Medical Private, Austrian Heart Foundation, University Burgenland (MPUB), Pinkafeld/Oberwart, Vienna, Austria; ¹⁵Department of Cardiology and Medical Intensive Care, Augustinerinnen Hospital Cologne, Cologne, Germany; ¹⁶Faculty of Medicine, University of Freiburg, Freiburg, Germany; ¹⁷Department of Cardiology, Rigshospitalet and Dept of Clinical Medicine, University of Copenhagen, Denmark; ¹⁸Emergency Institute for Cardiovascular Diseases “Prof. C.C. Iliescu”, Department 1; University of Medicine Carol Davila, Bucharest, Romania; ¹⁹Cardiology, IRCCS San Raffaele Scientific Institute, Milan, Italy; ²⁰Department of Cardiology, Vita-Salute San Raffaele University, Milan, Italy; ²¹Cardiology, ASST Spedali Civili, Brescia, Italy; ²²Department of Cardiology, Institute of Heart Diseases, Wroclaw Medical University, Wroclaw, Poland; ²³University of Zagreb School of Medicine, Department of Cardiovascular Diseases, University Hospital Centre Zagreb, Zagreb, Croatia; ²⁴Department of Cardiology, Transplant Center Utrecht, University Medical Center Utrecht and Utrecht University, Utrecht, The Netherlands; ²⁵Intensive Cardiac Care Unit, Cardiology Department, Hospital de la Santa Creu i Sant Pau, IIB-Sant Pau, Universitat Autònoma de Barcelona, CIBER-CV, Barcelona, Spain; ²⁶Torino North Emergency San Giovanni Bosco, Turin, Italy; ²⁷Institut de Cardiologie, Hôpital de la Pitié-Salpêtrière, APHP, Sorbonne Université, Paris, France; ²⁸Faculty of Medicine and Surgery, Kore University of Enna, Enna, Italy; ²⁹Policlinico Centro Cuore GB Morgagni, Catania, Italy; ³⁰National Heart and Lung Institute, Imperial College London, Dovehouse Street, London SW8 5NP, UK; ³¹School of Cardiovascular and Metabolic Medicine & Sciences, King's College London, James Black Centre, Denmark Hill Campus, 125 Coldharbour Lane, London SE 5 8NL, UK; ³²Center for Molecular Cardiology, University of Zürich, Wagistrasse 14, Schlieren—Zürich 8952, Switzerland; ³³Heart Division, Royal Brompton and Harefield Hospitals, Sydney Street, London SW3 5NP, UK; ³⁴Department of Cardiology, University Heart and Vascular Center Hamburg, University Medical Center Hamburg-Eppendorf, Martinistr. 52, 20251 Hamburg, Germany; ³⁵Centre for Population Health Innovation (POINT), University Heart and Vascular Center Hamburg, University Medical Center Hamburg-Eppendorf, Hamburg, Germany; and ³⁶German Center for Cardiovascular Research, Partner Site Hamburg/Kiel/Luebeck, Hamburg, Germany

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* Corresponding author. Tel: 0049 40 7410 0, Email: b.schrage@uke.de

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Cardiogenic shock is a complex clinical syndrome with a high mortality risk. As heterogeneity of patient presentation is high and diagnosis not standardized, a timely diagnosis and thus initiation of treatment are oftentimes not achieved, further worsening the prognosis of affected patients. We therefore propose standardized criteria for the diagnosis of cardiogenic shock, based on three essential pillars: (i) an evidence of end-organ hypoperfusion at rest, (ii) which is primarily caused by haemodynamic abnormalities and a direct consequence of (iii) an underlying cardiac dysfunction commensurate with the state of shock.

Introduction

Cardiogenic shock (CS) is a clinical syndrome. Under this umbrella of the syndrome of CS is the significant heterogeneity in clinical presentation, pathophysiology, clinical course, and outcomes. The heterogeneity of the CS syndrome complicates the conduct, interpretation, and comparison of clinical studies, not least because of a lack of standardized definitions of CS. This issue of standardized criteria for CS has assumed greater urgency as the number of registries and clinical trials.

With the support of the Association for Acute CardioVascular Care (ACVC) of the European Society of Cardiology (ESC), an initiative to review the criteria for the diagnosis of CS was established. This initiative was subsequently expanded in 2025 to include other stakeholders within the ESC, including the Heart Failure Association (HFA) and the European Association of Percutaneous Cardiovascular Interventions (EAPCI) to conceptualize the criteria for the definition of CS. The members of this group were nominated and approved by ACVC, HFA, and EAPCI.

What is cardiogenic shock?

Shock is generally defined as a state of circulatory failure that results in inadequate cellular oxygen (O_2) supply, which ultimately may lead to the development of end-organ injury and failure. In practice, cellular O_2 supply vs. demand cannot be directly assessed but can be inferred from clinical and/or metabolic changes associated with hypoperfusion. However, such clinical and/or metabolic parameters are not uniformly used in the literature or in clinical trials. Nevertheless, CS is a state of shock that is caused predominantly by a haemodynamic abnormality due to an underlying cardiac pathology. Thus, in essence, CS is the combination of the following:

- (1) Evidence of end-organ hypoperfusion at rest
- (2) Primarily caused by haemodynamic abnormalities and a direct consequence of
- (3) An underlying cardiac dysfunction commensurate with the state of shock

Challenges with the clinical diagnosis of cardiogenic shock

While this pathophysiologic definition of CS is not contentious, the translation of this descriptive definition into clinical practice has not been uniform, such that the criteria proposed by international societies and used by 'real-world' registries and clinical trials are not aligned [Table 1]. Indeed, some published and ongoing randomized clinical trials (RCTs) have not specified diagnostic criteria for CS, leaving the diagnosis to the clinician's discretion. In addition, the criteria are weighted towards acute myocardial infarction (AMI)-related CS. Despite the well-reported differences between AMI- and heart failure (HF)-related CS,¹⁸ the relative abundance of clinical trials in the former has shaped the diagnostic

criteria for CS, providing further challenges for clinicians and scientists working in this field alike.

Need for diagnostic criteria to improve clinical applicability

Diagnostic criteria are the description of an illness concept. Clinical criteria operationalize the definitions of an illness and are frequently used as inclusion criteria for RCTs. These, in turn, determine the heterogeneity of the subjects included into clinical studies and, by extension, the variability in the reported incidence and outcomes of the studies. Broad, non-specific criteria allow the erroneous inclusion of subjects without the condition of interest and confound the interpretation of the study results. For example, the patients included in the SHOCK (SHould we emergently revascularize Occluded Coronaries for cardiogenic shock) registry showed a wide range of haemodynamic parameters,¹⁹ including some patients with cardiac indices of >3 L/min/m² and some patients with isolated hypotension without clinical evidence of hypoperfusion. Not surprisingly, patients without hypoperfusion had significantly better short-term survival.²⁰ The criteria used by clinical trials may also differ because some trials deliberately target specific patient groups or severity. For example, ECLS-SHOCK targeted patients with more severe CS and used a lactate level of >3 mmol/L to include patients at higher risk of mortality.²¹ Additionally, some randomized trials may attempt to target specific subpopulations of patients; e.g. the investigators of the DanGer Shock trial made the deliberate choice to only enrol patients with a lower risk of non-cardiovascular death, to provide a better match between study population and tested intervention.¹³

Criteria that distinguish advanced HF from CS are especially relevant, as HF-related CS is increasingly prevalent.²² Haemodynamic abnormalities [e.g. low systolic blood pressure (SBP) and low cardiac index] are commonly used as criteria for CS, but these features are also common in patients with advanced HF²³ and in patients with decompensated HF without shock.²⁴ Indeed, in the ESCAPE trial, a large proportion of patients with advanced HF would meet the commonly used haemodynamic criteria for CS.²⁵ Similarly, liver and renal dysfunction, mainly attributable to congestion (i.e. cardiohepatic²⁶ and cardiorenal²⁷ syndrome), are common in advanced HF and impede a distinction between advanced HF and CS. Thus, haemodynamic abnormalities must be associated with evidence of resting hypoperfusion to meet the definition of CS. For example, the broad set of criteria for CS adopted by the Cardiogenic Shock Working Group [Table 1] would have allowed the inclusion of patients with advanced or decompensated HF without shock. In fact, this may have contributed to the description of the 'non-congested' and 'cardiorenal' phenotypes, which were predictably significantly more common in patients diagnosed with HF-CS compared to AMI-CS and often characterized by normal blood lactate levels.¹⁵ Not surprisingly, mortality rates were significantly lower in these two phenotypes compared to the 'cardiometabolic' phenotype, which was characterized by significantly higher arterial blood

Table 1 Criteria for cardiogenic shock from clinical guidelines, trials, and registries

Source	Criteria
Guidelines or statements	
AHA/ACC/HFSA Heart Failure Guidelines 2022 ¹	Hypotension: systolic blood pressure <90 mmHg for >30 min or mean blood pressure <60 mmHg for >30 min or vasopressors to maintain systolic blood pressure ≥90 mmHg or mean blood pressure ≥60 mmHg Hypoperfusion: Decreased mentation, cold extremities/livedo reticularis, urine output <30 mL/h, lactate >2 mmol/L
ESC Heart Failure Guidelines 2021 ²	Diagnosis of cardiogenic shock mandates the presence of clinical signs of hypoperfusion (cold extremities, oliguria, mental confusion, dizziness, narrow pulse pressure) Biochemical manifestations of hypoperfusion (elevated serum creatinine and lactate, metabolic acidosis). Hypoperfusion is not always accompanied by hypotension
HFA of the ESC Position Statement 2020 ³	A syndrome caused by a primary cardiovascular disorder in which inadequate CO results in a life-threatening state of tissue hypoperfusion associated with impairment of tissue O ₂ metabolism and hyperlactataemia which, depending on its severity, may result in multi-organ dysfunction and death
ACVC of the ESC position statement 2020 (acute myocardial infarction complicated by cardiogenic shock) ⁴	All of the following (A) Systolic blood pressure <90 mmHg for >30 min or need for vasopressors to maintain systolic pressure >90 mmHg (B) Clinical signs/symptoms At least one of the following (1) Altered mental status (2) Cold, clammy skin and extremities (3) Oliguria with urine output <30 mL/h (4) Arterial lactate >2.0 mmol/L And elevated left ventricular filling pressures Pulmonary congestion Elevated pulmonary artery wedge pressure (including Doppler echocardiography) Left ventricular end-diastolic pressure (LVEDP) at catheterization >20 mm Hg (C) Cause is cardiogenic LVEF <40% Mechanical complications (e.g. severe mitral regurgitation and septal rupture) Right ventricular failure or infarction
Shock Academic Research Consortium ⁵	'Pragmatic' definition: Cardiac disorder that results in both clinical and biochemical evidence of sustained tissue hypoperfusion 'Clinical trial' definition: Cardiac disorder that results in a systolic blood pressure <90 mm Hg for ≥30 min (or the need for vasopressors, inotropes, or mechanical circulatory support to maintain systolic blood pressure ≥90 mm Hg) with evidence of hypoperfusion Haemodynamic criteria (optional) Cardiac index <2.2 L/min/m ² Hypoperfusion (≥1 of the following): arterial lactate >2 mmol/L, acute kidney injury (creatinine ≥2× upper limit of normal) or oliguria (e.g. urine output <0.5 mL/(kg·h), acute hepatic injury (e.g. ALT >3× upper limit of normal), cool or mottled extremities, altered mental status not explained by an alternative cause Haemodynamic criteria (optional) Cardiac index <2.2 L/min/m ² Systemic vascular resistance >2200 dynes(cm.s ⁻⁵)
Clinical trials	
SHOCK ⁶	Acute myocardial infarction Clinical criteria: Hypotension (a systolic blood pressure of <90 mmHg for at least 30 min or the need for supportive measures to maintain a systolic blood pressure of ≥90 mmHg) and end-organ hypoperfusion (cool extremities or a urine output of <30 mL per hour, and a heart rate of ≥60 b.p.m.).

Continued

Table 1 Continued

Source	Criteria
TRIUMPH ⁷	Haemodynamic criteria: Cardiac index of <2.2 L/min/m² and pulmonary–capillary wedge pressure of ≥ 15 mmHg. Acute myocardial infarction Peripheral signs of tissue hypoperfusion and systolic blood pressure <100 mmHg despite vasopressor therapy (dopamine 7 µg/kg per minute or norepinephrine or epinephrine 0.15 mcg/kg/min) continuing longer than 1 h Clinical or haemodynamic evidence of elevated left ventricular filling pressures Left ventricular ejection fraction $<40\%$
CULPRIT-SHOCK ⁸	Acute myocardial infarction Systolic blood pressure <90 mmHg for more than 30 min or needed infusion of catecholamines to maintain a systolic pressure above 90 mmHg, Clinical signs of pulmonary congestion and had impaired end-organ perfusion. The diagnosis of impaired end-organ perfusion required at least one of the following: (a) Altered mental status (b) Cold, clammy skin and extremities (c) Oliguria with urine output <30 mL per hour (d) Serum lactate level >2.0 mmol/L.
IABP-SHOCK II ⁹	Acute myocardial infarction Systolic blood pressure <90 mmHg for more than 30 min or needed infusion of catecholamines to maintain a systolic pressure above 90 mmHg. Clinical signs of pulmonary congestion and had impaired end-organ perfusion. The diagnosis of impaired end-organ perfusion required at least one of the following: (a) Altered mental status (b) Cold, clammy skin and extremities (c) Oliguria with urine output <30 mL per hour (d) Serum lactate level >2.0 mmol/L
DOREMI ¹⁰	One of the following (A) Clinical diagnosis of CS and systolic blood pressure <90 mmHg with end-organ dysfunction (B) Clinical evidence of systemic and/or pulmonary congestion despite use of vasodilators and/or diuretics (C) Acute coronary syndrome complicated by cardiogenic shock with haemodynamic reduction in cardiac index (<1.8 L/min/m² and left ventricular end-diastolic pressure >18 mmHg) (D) A clinically determined need to augment cardiac output in addition to ongoing vasopressor therapy (E) A treating team's clinical assessment that inotropic therapy is required for developing cardiogenic shock without current evidence of hypoperfusion
EURO-SHOCK ¹¹	Acute myocardial infarction (I) Systolic blood pressure <90 mmHg for >30 min or catecholamines required to maintain systolic blood pressure >90 mmHg (II) Clinical signs of pulmonary congestion (III) Hypoperfusion with at least one of the following (a) Altered mental status (b) Cold and clammy skin and limbs (c) Oliguria (<30 mL/h) (d) Lactate >2 mmol/L
ECLS Shock ¹²	Acute myocardial infarction, plus obligatory (I) Planned revascularization (II) Systolic blood pressure <90 mmHg for >30 min or catecholamines required to maintain systolic blood pressure >90 mmHg

Continued

Table 1 Continued

Source	Criteria
	(III) Signs of impaired organ perfusion with at least one of the following criteria (a) Altered mental status (b) Cold, clammy skin and extremities (c) Oliguria with urine output <30 mL/h (IV) Arterial lactate >3 mmol/L
DanGer Shock ¹³	Acute myocardial infarction (a) Peripheral signs of tissue hypoperfusion (arterial blood lactate ≥ 2.5 mmol/L and/or SvO ₂ <55% with a normal PaO ₂) (b) Systolic blood pressure <100 mmHg and/or need for vasopressor therapy (dopamine/norepinephrine or epinephrine) (c) Left ventricular ejection fraction <45%
Altshock-2 ¹⁴	Acute decompensated heart failure and all of the following: • Need for vasoactive agents to maintain SBP >90 mmHg or MAP > 60 mmHg • Left ventricle systolic function $\leq 35\%$ • SCAI B-D according to CSWG definition Cardiogenic shock endpoint defined as Hypotension (felt to be secondary to cardiac dysfunction): systolic blood pressure <90 mmHg for >1 h not responsive to fluid resuscitation and/or heart rate correction or systolic blood pressure <90 mmHg and increases to ≥ 90 mmHg <1 h with positive inotropic or vasopressor agents alone and/or with mechanical support; Hypoperfusion (a) Cool, clammy skin (b) Oliguria (urine output <30 mL/h) (c) Altered sensorium (d) Cardiac index <2.2 L/min/m ²
Multi-centre registries	
CSWG ¹⁵	Any one of the following • Sustained episode of SBP ≤ 90 mmHg ≥ 30 min or need for vasoactive agents to maintain BP • Cardiac index <2.2 L/min/m ² due to cardiac dysfunction • Use of temporary mechanical circulatory support
CCCTN ¹⁶	• Sustained episode of SBP <90 mmHg for ≥ 30 min or need for vasoactive agents to maintain BP • Evidence of end-organ hypoperfusion (altered mental state, oliguria, acute kidney/hepatic injury, or lactate >2 mmol/L) • If available, cardiac index <1.8 or <2.2 L/min/m ² (on inotrope) with elevated filling pressures
FRENDSHOCK ¹⁷	At least one in each of the following • Systolic blood pressure <90 mmHg and/or the need for amines to maintain systolic blood pressure >90 mmHg and/or cardiac index <2.2 L/min/m ² on echocardiography or right heart catheterization • Elevation of left and/or right heart pressures, defined by clinical signs, radiology (overload signs on chest X-ray or computed tomography scan), biological tests (natriuretic peptide elevation), echocardiography (usual signs of left ventricular filling elevation), or invasive haemodynamic overload signs (elevation of mean pulmonary artery pressure or pulmonary capillary wedge pressure) • Signs of malperfusion, which could be clinical (oliguria, mottling, confusion) and/or biological (lactate >2 mmol/L, hepatic insufficiency, renal failure)

ACC, American College of Cardiology; AHA, American Heart Association; CCCTN, Critical Care Cardiology Trials Network; CSWG, Cardiogenic shock working group; CULPRIT-SHOCK, Culprit Lesion Only PCI Versus Multivessel PCI in Cardiogenic Shock; ESC HFA, European Society of Cardiology Heart Failure Association; ESC, European Society of Cardiology; HFSA, Heart Failure Society of America; IABP-SHOCK II, Intraaortic Balloon Pump in Cardiogenic Shock II; SHOCK, Should We Emergently Revascularize Occluded Coronaries for Cardiogenic Shock.

lactate levels compared to the other two phenotypes [lactate: 'cardiometabolic' 8.2 (5.4–11.4) vs. 'non-congestive' 1.8 (1.2–2.9) vs. 'cardiorenal' 2.0 (1.4–3.0) mmol/L, $P < 0.001$].²⁸

The overlap of advanced or decompensated HF with CS is not unique to the criteria adopted by the Cardiogenic Shock Working Group. The SHARC (Shock Academic Research Consortium)

criteria also allow for the inclusion of patients with HF, hypotension, and organ dysfunction. The results are similarly predictable—patients with CS due to ‘acute-on-chronic’ HF had the lowest (often normal) blood lactate levels [lactate: 2.2 (1.3–3.8) mmol/L] and the lowest in-hospital mortality.²⁹

Non-uniformity of criteria for CS complicates comparisons between studies and may compromise the external validity of RCTs.³⁰ For example, the DOREMI (Dobutamine Compared to Milrinone) trial reported no significant difference in in-hospital mortality between milrinone and dobutamine in a mixed group of patients with AMI- and non-AMI-CS.¹⁰ However, the criteria used by the DOREMI investigators were significantly broader than the criteria used in other clinical trials of AMI-CS. Using the relatively contemporary IABP-SHOCK II (Intraaortic Balloon Pump in Cardiogenic Shock) trial as a comparison,⁹ the patients with AMI-CS in DOREMI had higher median blood pressures [systolic: 108 [interquartile range (IQR) 98–118] vs. 89 [IQR 79–107] mmHg; diastolic: 59 [IQR 54–70] vs. 55 [IQR 46–67] mmHg; mean: 75 [IQR 67–84] vs. 69 [IQR 59–80] mmHg] and lower blood lactate levels [2.8 (IQR 1.7–4.3) vs. 3.6 (IQR 2.1–7.2) mmol/L].³¹ These differences may be clinically significant,³² and it is unclear if the results of the DOREMI trial can be extrapolated to other patients with AMI-CS.

Diagnostic criteria for cardiogenic shock—European perspective

There is an important need for features that can be easily identified and measured in individual patients to increase uniformity. Ideally, these clinical criteria should identify all three diagnostic components of CS (e.g. hypoperfusion, haemodynamic abnormality, and cardiac origin), be easily obtainable in clinical practice, and be available promptly and at an acceptable cost. In addition, they should be applicable in various clinical settings from out-of-hospital, emergency department/shock room, to hospital ward or intensive care settings to allow early recognition and triage of patients at risk of progressing to or at a life-threatening state. Early recognition is particularly important because prompt and specific therapy, such as mechanical circulatory support, may improve outcome but likely only in well-selected patients with ‘true’ CS. Furthermore, it should be possible to test the validity of these criteria in prospective studies. Crucially, the criteria for CS must include all three components of CS.

(A) Evidence of end-organ hypoperfusion at rest

Circulatory failure results in inadequate O₂ delivery to parenchymal cells. In response, regional vascular beds vasoconstrict to preferentially divert blood flow to vital organs. Vasoconstriction of some vascular beds such as the skin manifests as cold peripheries and mottling. Regional redistribution of blood flow is accompanied by a compensatory increase in O₂ extraction, which lowers the O₂ content of mixed venous blood and mixed venous O₂ saturation. Anaerobic metabolism, hyperlactataemia, and organ dysfunction ensue when O₂ extraction mechanisms are exhausted. Early correction of systemic O₂ delivery variables can rapidly reverse the clinical signs of hypoperfusion. However, in advanced or late CS, cellular hypoxia and reduced oxygen extraction may persist despite restoration of oxygen delivery, due to failure of the microcirculation and mitochondria to transport and efficiently use oxygen. This dissociation of systemic O₂ delivery variables and the microcirculation has been described as the loss of haemodynamic coherence.³³ The failure of cellular O₂ utilization or cellular

dysoxia may paradoxically result in increased mixed venous O₂ saturation but persistent organ dysfunction.

Circulatory abnormalities such as hypotension and low cardiac index are common to both advanced HF and CS. However, unlike HF, CS is characterized by the failure of oxygen delivery below the critical point of supply dependence, resulting in inadequate tissue perfusion and oxygenation at rest. Resting hypoperfusion discriminates CS from HF.

Vincent³⁴ described the three windows into the body—skin, brain, and kidneys—for the clinical assessment of perfusion. Metabolic parameters, most notably blood lactate level, have also been used as markers of hypoperfusion. There are limitations to each of these clinical parameters [Table 2]. In general, clinical features of hypoperfusion (i) have a modest relationship with macrocirculatory abnormalities such as arterial blood pressure or cardiac index and (ii) more closely reflect microcirculatory abnormalities; (iii) their severity is associated with short-term mortality; and (iv) early resolution of those on serial assessment is associated with better clinical outcomes. Improvement in global haemodynamic parameters (i.e. macrocirculation) with persistent hyperlactataemia and organ dysfunction would suggest the loss of haemodynamic coherence (see above).

There are other methods of assessing the microcirculation and perfusion, such as the use of hand-held vital microscopy (e.g. sidestream dark-field and incident dark-field imaging techniques) to quantify sublingual microcirculatory density and per cent perfused vessels; but these techniques have yet to enter clinical use. Of note, one medium-sized RCT showed that treatment changes in response to sublingual microcirculatory perfusion variables did not improve survival.⁴⁴

In the absence of a single ‘ideal’ measure of tissue perfusion, a multi-parametric approach to the assessment of perfusion would seem appropriate. This multi-parametric approach would consider a combination, but not necessarily all, of the following parameters, so that it can be tailored to the given clinical setting:

- (i) Clinical markers of peripheral perfusion [e.g. mottling presence and extension,⁴⁵ skin temperature, and capillary refill time can reliably identify patients at higher risk (Table 2)].⁴⁶
- (ii) Objective point-of-care measurements (e.g. blood lactate level). Changes in blood lactate concentrations are also important with a greater prognostic value of absolute blood lactate concentrations at 24 h than on admission, as shown in a prospective observational registry.⁴⁷ Spontaneous hypoglycaemia may develop in cases of significant ischaemic liver injury; in one study of patients with ischaemic hepatitis, hypoglycaemia was noted in 32.4% of cases.⁴⁸
- (iii) More complex and invasive bedside objective measurements (e.g. arterial and central or mixed venous O₂ saturation and partial pressure of carbon dioxide gradients may be prognostically significant)⁴⁹ and near-infrared spectroscopy
- (iv) Objective laboratory parameters of organ dysfunction (e.g. rise in creatinine, liver transaminases, bilirubin, and thrombocytopenia).

B. Haemodynamic abnormality

Low cardiac index is *conditio sine qua non* for CS. By definition, cardiac filling pressures must be normal or elevated (i.e. not related to hypovolemia). Low cardiac index in CS is frequently, but not universally, accompanied by systemic arterial hypotension. Nonetheless, systemic arterial hypotension is used in most of the proposed criteria for CS.

The definition of systemic arterial hypotension in circulatory shock varies. Many clinical trials of AMI-related CS have used SBP <90 mmHg, which has led to the widespread use of this cut-off to define hypotension in CS. However, mean arterial blood

Table 2 Markers of hypoperfusion for the diagnosis of cardiogenic shock

Parameter	Comments
Clinical assessment	
Cold, clammy skin	Used in clinical trials, but subjective and poorly defined. Not specific to shock states
Capillary refill time (CRT)	In cardiogenic shock, a CRT of >3 s was associated with mechanical circulatory support and 90-day mortality. ³⁵ Poor correlation with macrocirculatory parameters, such as blood pressure and cardiac index. Associated with reduction in lactate level. Reflects microcirculatory abnormalities. Affected by skin temperature (longer in hypothermia).
Skin mottling	A 6-degree scale to quantify skin mottling: 0 = no mottling; 1 = small mottling area (coin size) at the centre of the knee; 2 = mottling area that does not exceed the superior edge of the kneecap; 3 = mottling area that does not exceed the middle thigh; 4 = mottling area that does not go beyond the fold of the groin; 5 = extremely severe mottling area that goes beyond the fold of the groin. A higher grade is associated with mortality in critically ill patients, ³⁶ with higher mortality in grade ≥ 2 . ³⁷ Limited data in cardiogenic shock. Poor correlation with macrocirculatory parameters, reflects microcirculatory abnormalities
New altered mental state	Not specific to shock. May be affected by medications (e.g. benzodiazepines and opiates) and intracranial pathology. Associated with higher mortality if altered mental state is related to cardiogenic shock ³⁸
Oliguria	Urine output <0.5 mL/kg/h. Indicative of renal impairment, not specific for cardiogenic shock. Delayed marker. Requires catheterization for accurate quantification
Rapid point-of-care measurement	
Blood lactate	Levels of >2 mmol/L are generally considered abnormal, and persistent hyperlactataemia is consistent with cardiogenic shock. Higher level associated with higher mortality. Rapid reduction in lactate associated with better outcomes (termed 'lactate clearance'). Poor lactate clearance despite improvement in haemodynamic parameters may indicate organ ischaemia/infarction, microcirculatory dysfunction, and/or cellular dysoxia (loss of haemodynamic coherence). Not specific to hypoperfusion; other causes include aerobic glycolysis (e.g.: epinephrine) and drugs (e.g.: metformin)
Base excess	Deficits (negative) in base excess indicative of renal dysfunction. Renal dysfunction not specific to cardiogenic shock. May be affected by medications (e.g. furosemide).
Near-infrared spectroscopy (NIRS)	Based on differential absorption spectrums of oxygenated and deoxygenated haemoglobin, in small, mostly venous vessels (<1 mm). Can be affected by skin tone. ³⁹ Tissue haemoglobin O ₂ saturation (StO ₂) can be calculated, which represents the local balance between O ₂ delivery and consumption. NIRS-derived StO ₂ can predict mortality in circulatory shock, but the impact of NIRS monitoring on outcomes is uncertain. Standardization of methodology and clinical randomized trials are needed before wider clinical use
Invasive bedside assessment	
Mixed venous oxygen saturation (SvO ₂)	Typically measured from pulmonary artery. Related to O ₂ delivery and consumption (lower at higher levels of oxygen extraction, due to inadequate oxygen delivery and/or increased oxygen consumption), not hypoperfusion <i>per se</i> . There is not a single SvO ₂ cut-off to define hypoperfusion in cardiogenic shock, as low SvO ₂ can be tolerated in some patients with advanced HF without CS. ⁴⁰ Central and SvO ₂ in cardiogenic shock due to myocardial infarction are typically <50–55% and lower in decompensated chronic HF. ^{41–43}

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Table 2 Continued

Parameter	Comments
	May paradoxically be normal (>70–75%) or supranormal in severe cardiogenic shock due to microcirculatory and/or cellular dysoxia.
	Serial assessment may identify deteriorating cardiogenic shock.
Venous–arterial gradient in the partial pressure of CO ₂ ($\Delta P_{\text{v-aCO}_2}$)	Small gradient normally (<6 mmHg or 0.8 kPa). High gradient in low flow state (low cardiac output) and not hypoperfusion <i>per se</i> . Minor measurement error will have significant impact. Requires central venous cannulation.
Ratio of $\Delta P_{\text{v-aCO}_2}$ to arterial–venous oxygen content ($\Delta P_{\text{v-aCO}_2}$: ΔCavO_2 ratio)	Ratio of >1.4 mmHg.dL/mL O ₂ indicative of anaerobic metabolism. Cumbersome calculation. Little data in cardiogenic shock. Requires central venous/pulmonary artery catheterization
Organ dysfunction	
Creatinine	Not immediately available and non-linear relationship with glomerular filtration rate. Usually delayed/late marker.
Liver transaminase and bilirubin	Not always caused by low cardiac output. Kinetics of transaminase more rapid than bilirubin (bilirubin may lag behind by 48 h).
Platelet count	Thrombocytopenia is common, especially with mechanical circulatory support, reflects severity of shock.

pressure (MAP) is also frequently used, with a MAP <60 mmHg used interchangeably with SBP <90 mmHg to define CS.⁵⁰ Physiologically, the relationship between SBP and MAP is weak. Diastolic blood pressure (DBP) is the dominant determinant of MAP, and related to arterial resistance, while age-related changes in aortic compliance greatly influence SBP.⁵¹ As such, a patient may have a MAP <60–65 mmHg, without being hypotensive according to the SBP criterion of <90 mmHg, and vice versa. Data from the Mayo Clinic cardiac intensive care unit showed that 26.4% of patients had low SBP (<90 mmHg), 26.2% had low MAP (<60 mmHg), and only 47.3% had hypotension by both criteria.⁵⁰ Mortality was highest in patients who met both SBP and MAP criteria for hypotension. Numerous studies have confirmed the association between low SBP and MAP with higher mortality in CS, even if the specific threshold differed between studies.^{49,52} Other blood pressure-derived parameters such as organ perfusion pressure (difference between MAP and central venous pressure)⁵³ and diastolic perfusion pressure (difference between DBP and central venous pressure)⁵⁴ may also be associated with mortality in CS.

This dilemma of whether to use SBP or MAP for the diagnosis of CS cannot be reconciled, due to the inherent physiologic limitations of arterial BP as a parameter. Similarly, there are inevitable limitations in any single cut-off for SBP or MAP. These limitations simply reflect the fact that the complexities of circulatory shock cannot be captured by a single parameter and certainly not by a single value.⁵⁵ Therefore, arterial BP, be it SBP or MAP, must be interpreted in conjunction with other haemodynamic parameters and evidence of hypoperfusion.

It is also important to recognize that arterial BP is not synonymous with tissue perfusion. This is especially pertinent in so-called normotensive CS, which is defined as CS with a SBP >90 mmHg. The relatively preserved SBP resulting from more severe and indiscriminate vasoconstriction in response to low cardiac output (higher systemic vascular resistance) is associated with organ hypoperfusion and short-term prognosis comparable to patients with hypotensive CS.²⁰

Congestion, like hypotension, is common but not a universal feature of CS. Congestion can be evaluated clinically, such as

the jugular venous pressure, lung crackles, or third heart sound, on chest radiograph, by echocardiography, lung ultrasound, or direct haemodynamic assessment from cardiac catheterization (haemodynamic congestion). In one series, 18% of patients with apparent CS had right atrial and pulmonary artery wedge pressures of <12 mmHg and <18 mmHg, respectively⁵⁶—a group of patients described as ‘euvolaemic’ CS.

Organ or local blood flow is dependent on local perfusion pressure and vascular resistance and congestion. Vasodilatation lowers resistance to flow due to reduced small arterial/microvascular resistances and increases blood flow—this lowers the systemic arterial BP required for autoregulation (lower limit of autoregulation). The ability to increase blood flow with vasodilatation, i.e. the flow reserve, varies between organs. In a dog animal study, Magder et al. demonstrated more limited flow reserve in the kidneys and higher pressure at zero-flow despite vasodilatation with nitroprusside compared to the heart.⁵⁷ This phenomenon may explain why the kidneys are effectively the ‘canaries in the coal mine’ of arterial BP, with renal injury occurring early, even before evidence of cardiac or cerebral hypoperfusion. However, some features of renal injury, such as oliguria and a rise in serum creatinine, take time to manifest and may not be detectable at the onset of renal injury. Of note, the pressure at zero-flow that defines the limit of autoregulation is lowered with vasodilatation and increased with vasoconstriction and pathological ‘obstructive’ conditions such as tamponade or intra-abdominal hypertension.

Cardiac indices (CI) of <1.8–2.2 L/min/m² have been used as a haemodynamic criterion for CS. Unlike arterial BP, CI is difficult to measure in practice. Cardiac index measurements derived from a pulmonary artery catheter (PAC), such as thermodilution or indirect Fick method, are not widely available. Thus, a PAC-derived CI to diagnose CS would not be practical. In many cases, low CI would need to be inferred from echocardiographic findings [e.g. low velocity time integral (VTI)], calculated stroke volume and CI (from aortic VTI, sub-aortic diameter, and heart rate), or LV ejection fraction (LVEF) relative to LV volume. In general, with the exception of rare cases of high output HF (e.g. chronic liver disease or concomitant sepsis) and obstructive

shock, the co-existence of low arterial BP and elevated cardiac filling pressures (e.g. high central venous pressure/distended inferior vena cava or high pulmonary artery wedge pressure/restrictive transmitral Doppler pattern) is almost always attributable to low CI due to contractile dysfunction⁵⁸ (down- and rightward shift of the cardiac function curve).

It is generally believed that patients with chronic HF, unlike acute CS in AMI, have greater tolerance of lower O₂ delivery and arterial BP due to the development of adaptive mechanisms related to the exposure to chronic low-output state. However, these adaptive mechanisms have not been well-characterized. A rightward shift in the O₂ dissociation curve that improves O₂ delivery is one reported difference between patients with CS due to acutely decompensated chronic HF and AMI.⁴³ Chronic vasodilator therapy in HF may also lower the limit of BP autoregulation, allowing organ perfusion to be maintained at a lower arterial BP. Some studies suggest lower mortality in patients with CS due to acutely decompensated chronic HF compared to AMI,⁵⁹ but these studies are difficult to interpret because the diagnostic criteria that were used (SBP < 90 mmHg and organ dysfunction) may not adequately discriminate CS from advanced HF without shock, further emphasizing the importance of the criteria for CS before embarking on comparative studies.

C. Cardiac dysfunction

Knowledge of the aetiology of the underlying cardiac condition shapes our approach to a more specific treatment of CS. The aetiology may be evident in some cases based on clinical history alone (e.g. presentation with AMI or a prior diagnosis of cardiomyopathy and/or HF), but it is unclear in many cases with *de novo* presentation and often not adequately elaborated. The past medical history is also relevant for the assessment of the need for and the choice of mechanical circulatory support (e.g. prior functional status, frailty, severe peripheral vascular disease, previous mechanical aortic valve replacement, among others) and further clinical management (e.g. metabolic control in diabetes and pre-existing renal impairment). Electrocardiogram (ECG) and echocardiography are the main initial investigative tools, firstly to establish the underlying diagnosis and, secondly, to corroborate that the underlying cardiopulmonary pathology is commensurate with CS.

An ECG is widely available and should be part of the routine assessment in all patients with haemodynamic instability. Previous AMI or features consistent with cardiomyopathies and/or HF may provide diagnostic clues. Tachy- and brady-arrhythmias may require specific treatment.

LV ejection fraction is one of the most used metrics to describe cardiac function. Patients with HF are categorized based on the LVEF into reduced, mildly reduced, and preserved LVEF subgroups, and this practice has transcended HF practice into routine clinical use, including CS. However, the interpretation of LVEF is complex, especially in shock and/or in the presence of mitral regurgitation.⁶⁰ Firstly, this measure is highly pre- and afterload-dependent.⁶¹ Secondly, LVEF cannot reliably characterize haemodynamic status (e.g. normal LVEF does not exclude CS and low LVEF may be compatible with normal cardiac output and filling pressure). In the SHOCK trial, nearly 25% of patients had an LVEF of >40%.⁶² Thus, an abnormal LVEF may identify an underlying cardiac dysfunction, but the assessment of underlying cardiac pathology must extend beyond the simple measurement of LVEF. While the initial focussed echocardiographic assessment may be limited to an assessment of LVEF, a follow-up echocardiographic assessment for more detailed assessment should be considered to assess the following:

- (i) LV volume: Low LVEF in a normal or small LV volume is indicative of low stroke volume. Normal or small LV volume may be encountered in acute fulminant myocarditis, a condition characterized by low cardiac output and a highly malignant clinical course. In contrast, an LV that is 'severely impaired' with LVEF of 30% would produce a stroke volume of 60 mL if the LV is dilated with an end-diastolic volume of 180 mL. Studies have consistently reported lower LVEF in CS due to acutely decompensated HF in the context of AMI-CS as well as due to LV remodelling in chronic HF.¹⁸ The LV outflow tract VTI can also be measured to estimate the LV stroke volume and corroborate the estimation derived from the LVEF. Changes in CI may also be tracked by serial changes in LV outflow tract VTI. Thus, in addition to the assessment of structural abnormalities, haemodynamic evaluation should form part of the echocardiographic assessment of the LV in CS.
- (ii) Assessment of filling pressure: Transmitral Doppler is a well-established parameter for the assessment of LV diastolic filling properties. Raised LV end-diastolic pressure in the presence of normal LV size suggests an abnormal LV end-diastolic pressure-volume relationship (e.g. LV stiffness). A normal transmitral Doppler filling pattern is unusual in CS related to LV failure, especially in patients with reduced LVEF,⁶³ but Doppler assessment can be challenging in some cases due to body positioning, tachyarrhythmias, or pseudo-normalization (see⁶⁴). Similarly, normal central venous pressure or inferior vena cava dimensions are unusual in HF-related CS as right ventricular dysfunction and systemic congestion are generally present. However, inferior vena cava dimension may be difficult to interpret in some conditions (e.g. inspiratory efforts by the patient during mechanical ventilation, heavy 'abdominal breathing' during forced expiration in respiratory distress, compression of the vena cava or migration of the imaging plane related to excessive lateral movement of the vena cava). 'Euvoalaemic' CS has been reported in a minority of patients with CS, but the absence of elevated right- or left-sided filling pressures in circulatory failure should also prompt questions on the dominant shock pathophysiology (e.g. predominantly distributive).
- (iii) Other structural and/or functional abnormalities: In AMI, regional wall motion abnormalities may provide supportive diagnostic information, and mechanical complications should be excluded. Thickening of ventricular walls may be indicative of acute fulminant myocarditis with myocardial oedema formation. In end-stage cardiomyopathy, functional mitral and tricuspid valve regurgitation is common and may contribute to haemodynamic compromise. Primary acute valvular disease (e.g. acute papillary muscle rupture) may produce circulatory failure due to severe valvular regurgitation even in the presence of normal LVEF.⁶⁵ In hypertrophic cardiomyopathy or Takotsubo syndrome, LV impairment with concomitant functional LV outflow tract obstruction may guide decision-making towards the use of an implantable micro-axial flow pump, rather than vasopressor therapy.
- (iv) The right heart⁶⁶: Assessment of the right heart is mandatory, even if significant left heart pathology has been identified. Indeed, right HF is commonly associated with left heart disease,⁶⁷ and the clinical course of patients with HF with reduced LVEF deteriorates markedly with the development of right HF.⁶⁸ Indeed, in CS, concomitant right HF is associated with higher in-hospital mortality.⁶⁹ Cardiogenic shock due to primary acute right HF such as acute pulmonary embolism or acute *cor pulmonale* in severe acute respiratory failure is typically associated with right ventricular dilatation, septal shift, and raised central venous pressure. The ratio of tricuspid annular plane systolic excursion to systolic pulmonary artery pressure (TAPSE/SPAP ratio) is a surrogate of right ventricular-pulmonary arterial coupling, and a ratio of <0.31–0.33 is associated with higher mortality in patients with HF⁷⁰ and pulmonary hypertension,⁷¹ albeit the TAPSE/SPAP ratio has not been well studied in CS.

In summary, the demonstration of reduced LVEF alone is not sufficient. Echocardiographic assessment should (i) identify an

Table 3 Proposed criteria for cardiogenic shock

Criteria	
Cardiac dysfunction	Evidence of severe left, right, or biventricular dysfunction, and/or severe valvular disease, and/or mechanical complication. Recurrent/persistent ventricular arrhythmias Exclusion of obstructive shock (e.g. cardiac tamponade)
Haemodynamic abnormality	At least two of the following three criteria (I) SBP <90 mmHg and/or MAP <65 mmHg for >30 min or SBP ≥90 mmHg and/or MAP ≥65 mmHg on vasoactive drugs (II) Raised filling pressures either on clinical examination (e.g.: elevated jugular venous pressure and/or pulmonary oedema), imaging (e.g. lung ultrasound or chest radiograph), or direct measurement (CVP >12 mmHg and/or PAWP >18 mmHg) (III) Low cardiac index <2.2 L/min/m ² without vasoactive drugs (e.g. echocardiogram or pulmonary artery catheter)
Hypoperfusion	At least one clinical sign • Skin changes (skin changes: cold/clammy, mottling grade ≥2, and/or CRT >3 s) • New altered mental state/agitation • ^a Oliguria <0.5 mL/kg/h And at least one of the following metabolic parameters • Lactate >2.0 mmol/L • Low central or mixed venous oxygen saturation (typically <55%) • ^a Base excess −2 or lower • ^a Raised/rising creatinine >1.5 × from baseline or >26.5 μmol/L (0.3 mg/dL) increase • Raised/rising transaminase >2 × normal • Spontaneous hypoglycaemia unrelated to hypoglycaemic agents • ΔPvaCO ₂ > 6 mmHg or 0.8 kPa • ΔPvaCO ₂ : ΔCavO ₂ ratio >1.4 mmHg.dL/mL O ₂

For the diagnosis of CS, all three criteria—cardiac dysfunction, haemodynamic abnormality, and evidence of hypoperfusion—must be met with the combination of clinical and point-of-care assessments. Cardiac dysfunction may be evidenced by findings from clinical, electrocardiographic, or imaging assessment. Haemodynamic abnormality criteria allow for the diagnosis of ‘normotensive’ and ‘euvoalaemic’ CS. Cardiac index may be measured by echocardiography or by the use of a pulmonary artery catheter. Criteria for hypoperfusion must include a clinical sign to reflect clinical assessment of shock.

^aRenal dysfunction is associated with oliguria, negative base excess, and abnormal creatinine—only one of these three should be used. The creatinine increase is based on the KDIGO criteria for acute kidney injury.

underlying left and/or right heart or other structural/functional pathology and (ii) show that these abnormalities are associated with reduced flow (as a reflection of a low CI) and/or elevated filling pressures, to establish that the underlying cardiopulmonary pathology is compatible with CS. In support of such an approach, Mele *et al.* showed that echocardiography-derived profiling of LV forward flow (SV index on LV outflow tract VTI <30 mL/m², filling pressure (E/A ratio and inferior vena cava diameter >2.1 cm), and right ventricular function (TAPSE <17 mm and SPAP >35 mmHg) allowed for the categorization of patients hospitalized with decompensated HF and predicted all-cause mortality independently of LVEF.⁷² In skilled hands, echocardiography can be used to estimate cardiac output⁷³ and systemic vascular resistance⁷⁴ in CS, but such expertise in echocardiography may not be widely available. Imaging with transoesophageal echocardiography may be indicated in some cases of diagnostic uncertainty.

An inserted PAC allows for the direct measurement of several key haemodynamic parameters, as well as calculation of several additional parameters, and therefore provides haemodynamic insights in circulatory shock that can shape clinical management in CS, such as the following:

- In cases of diagnostic uncertainty, data from PAC can be used to confirm the diagnosis of CS, differentiate non-CS from CS, and

identify concomitant shock pathophysiology (e.g. concomitant vasodilatory or mixed shock).

- PAC-derived haemodynamic data improve characterization of left and right heart function (e.g. left- and right-sided filling pressures and pulmonary and systemic vascular resistance), improve phenotyping of CS, and identify the ‘true origin’ of shock (e.g. determining if CS is primarily caused by left, right, or biventricular HF).
- The ‘haemodynamic trend’ from serial PAC assessment may identify response to initial therapy, especially in cases of unexpected deterioration and guide interventions, including the use/escalation of mechanical circulatory support, although definitive evidence on this is still lacking.
- PAC-derived haemodynamic data are routinely used to assess candidacy for heart transplantation.

However, the interpretation of PAC-derived haemodynamic data has to be performed in the wider clinical context (e.g. valvular disorders, pre-existing HF) and requires caution, as several factors influence the actual measurements (e.g. mechanical ventilation). Importantly, the interpretation of PAC-derived haemodynamic values is closely linked to the correct interpretation of PAC waveforms. At present, there is wide variation in the use of PAC in CS that is likely related to diverging institutional practices.¹⁷ Randomized clinical trials (PACCS; Pulmonary Artery Catheter in Cardiogenic Shock trial, NCT05485376) may establish the role of PAC in HF-CS.

	End-organ hypoperfusion at rest	Hemodynamic abnormality	Cardiac dysfunction
Pre-hospital (suggested CS)	Skin mottling Prolonged capillary refill New or worsening mentation	Persistent (>30 minutes) SBP <90mmHg or MAP <65mmHg, or vasoactive drugs to maintain BP	Signs/symptoms of heart failure Medical history of cardiac disease Arrhythmias
Hospital ward (likely CS)	Oliguria Elevated lactate >2 mmol/L Base excess ≤ -2 Spontaneous hypoglycaemia	Estimated low cardiac output with normal or elevated filling pressure on focused echocardiography	Structural abnormalities on focused echocardiography or other cardiovascular imaging
Intensive care unit (confirmed CS)	↓ cvSO ₂ or mvSO ₂ ↑ Urea and creatinine ↑ Transaminase and/or bilirubin	Cardiac index <2.2L/min/m ² with normal or elevated filling pressures from invasive haemodynamics or in-depth echocardiography	Structural abnormalities on in-depth echocardiography

Figure 1 Proposed criteria for CS. For the diagnosis of cardiogenic shock, all three criteria—cardiac dysfunction, haemodynamic abnormality, and evidence of hypoperfusion—must be met with the combination of clinical and point-of-care assessments. The haemodynamic criteria allow for the diagnosis of ‘normotensive’ and ‘euvoalaemic’ CS. Cardiac index may be measured by echocardiography or using a pulmonary artery catheter, and the threshold of 2.2 L/min/m² applies in the absence of vasoactive drug therapy. Criteria for hypoperfusion must include a clinical sign. Renal dysfunction is associated with oliguria, negative base excess, and abnormal creatinine—only one of these three should be used. The creatinine increase is based on the KDIGO criteria for acute kidney injury. cvSO₂: central-venous oxygen saturation; mvSO₂: mixed venous oxygen saturation; SBP: systolic blood pressure; MAP: mean arterial blood pressure; BP: blood pressure. Hospital ward indicates clinical setting outside of the intensive care unit.

European proposal for a standardized diagnosis of cardiogenic shock

The clinical criteria for the diagnosis of CS should reflect the three components of the pathophysiologic CS definition [Table 3]. In the absence of an ideal parameter, multi-parametric criteria offer a pragmatic and practical diagnostic approach for CS that is also congruent with the underlying pathophysiology [Figure 1]. The proposed criteria share some similarities with the recent FRENDSHOCK criteria.⁷⁵ The greater emphasis on resting hypoperfusion in the proposed criteria aims to discriminate HF-CS from advanced or decompensated HF. The possibility of diagnosing CS without mandating hypotension (SBP <90 mmHg) or elevated filling pressures is deliberate, as ‘normotensive’ and ‘euvoalaemic’ CS are well-recognized but under-represented in clinical studies and continue to be excluded by recently proposed criteria for clinical trials. The criteria do not mandate any specific diagnostic modality but offer clinicians a range of parameters to establish the diagnosis of CS using various non-invasive or invasive diagnostic modalities, depending on availability at different clinical stages (e.g. pre-hospital vs. emergency room vs. intensive care unit). Recent evidence suggests a positive impact of PAC measurements on treatment decisions⁷⁶ and outcomes.⁷⁷ But until the ongoing randomized PACCS trial (NCT05485376), which tests the efficacy of PAC to improve outcomes in CS, provides further evidence on this, PAC use should specifically be

considered in complex or clinically unclear scenarios. The proposed schema does not mandate that criteria for all three components be met simultaneously, recognizing that clinicians often incorporate longitudinal data and evolving clinical, haemodynamic, or metabolic picture over time to establish the diagnosis [Figure 2]. In addition, the criteria acknowledge that invasive diagnostic procedures may be needed in some cases of greater complexity or diagnostic uncertainty [Figure 2]. Ultimately, this perspective may be the first step towards a *Universal Definition of Cardiogenic Shock*, comparable to the *Universal Definition of Myocardial Infarction*,⁷⁸ which has significantly harmonized scientific and clinical efforts in the field of myocardial infarction. Indeed, the considerations of this European group of experts strongly suggest that such a document is an unmet need for the field of CS, including clinical practice as well as research.

Discussion and future directions

The Society for Cardiovascular Angiography and Intervention (SCAI) published a staging scheme for CS in 2019⁷⁹ that takes into account that CS is a highly dynamic cardiac condition. It is therefore tempting to defer the diagnosis of CS to the SCAI staging criteria, which will obviate the need for any clinical criteria for CS. However, the SCAI criteria are, by their very design, subjective. Indeed, in their first publication, the SCAI writing group explicitly avoided objective parameters:

Example 1

Diagnosis of cardiogenic shock in the catheterization laboratory using clinical assessment, point-of-care markers and invasive assessment in a patient with acute myocardial infarction

History No known medical history Dyspnoea, abdominal pain Cold, clammy, mottling BP 90/63mmHg Sinus tachycardia JVP not seen Chest clear	Point-of-Care Lactate 3.5 Base excess -2 ECG: widespread ST elevation	Bedside Imaging LVEF 20%, normal LV volume LV wall thickening Pleural effusion (R > L)
	Laboratory Data Transaminase 1400U/L Bilirubin 48umol/L Creatinine 160umol/L	Invasive Assessment Central venous pressure 18mmHg Central venous O ₂ 52% Central venous Δ PvCO ₂ 10mmHg

Example 2

Diagnosis of cardiogenic shock in the intensive care unit using clinical assessment, point-of-care markers and longitudinal assessment in a patient with heart failure

History Recent myocardial infarction Dyspnoea Agitated Cold, clammy, mottling Pulmonary oedema BP 98/60mmHg Sinus tachycardia	Point-of-Care Lactate 6.0 Base excess -7 ECG: widespread Q waves	Bedside Imaging Echocardiogram: Severe LV dysfunction, SV 30ml Chest radiograph: Pulmonary oedema
	Longitudinal Data Intubated Norepinephrine for low BP	

Figure 2 Exemplified clinical application of the proposed diagnostic criteria of cardiogenic shock—European perspective. JVP: jugular vein pulse; ECG: electrocardiogram; LVEF: left ventricular ejection fraction; LV: left ventricle; BP: blood pressure.

'The authors recognize that the proposed classification scheme presented (like most) is arbitrary and fairly simple. Some may wish for stricter definitions of stages, or to tie stages to laboratory values or some kind of a scoring mechanism. However, we feel that the elegance of the classification resides with its simplicity...'

Ironically, there are now increasing calls for specific objective criteria to define the stages and severity of CS,⁸⁰ but these proposed criteria for SCAI staging seek to quantify the severity (e.g. number of vasoactive drugs or mechanical circulatory support), not to diagnose CS. Staging systems for an illness follow the diagnosis of the illness. Criteria for diagnosis and staging should not be conflated.

The proposed criteria for CS do not obviate the need for a staging system. Quite the contrary, the standardization of CS criteria would complement the SCAI staging schema. The application of the SCAI CS staging scheme to an inconsistently defined cohort of patients with circulatory abnormality could result in highly variable patient characteristics and outcomes that would hamper clinical interpretation of clinical studies. This was evident from the 'validation studies' summarized in the follow-up consensus update in 2022.⁸¹ The individual 'validation studies' had different criteria for the diagnosis of CS.^{82–84} For the same SCAI stage, there was almost five-fold difference in short-term mortality (from <10%–

>50% in SCAI stage C CS), and in several studies, the mortality in SCAI stage C was even comparable to SCAI stage E. Arguably, the application of the SCAI staging schema to more uniformly defined cohorts will produce more uniform outcomes between studies for the same SCAI stage, improving communication between clinicians and the conduct of clinical trials.

Limitations

Several subject areas have been deliberately excluded as they are beyond the scope of this document. Firstly, increasing recognition of the heterogeneity in CS has led to studies of CS phenotypes.²⁸ Standardized diagnosis of CS may facilitate future studies to better characterize CS phenotypes. Secondly, the terms vasoplegia and distributive or 'mixed' shock are often used interchangeably to describe CS complicated by abnormal vasodilatation. Some have proposed a framework for 'mixed' shock,⁸⁵ but more studies are needed to better characterize this condition. Thirdly, the term 'refractory shock' is commonly used, especially in the context of mechanical circulatory support and may require further study. Indeed, the criteria used by clinical trials of mechanical circulatory support differed from other trials in CS to identify 'sicker' patients who may be refractory to pharmacological therapy.¹² A set of criteria for refractory shock is especially relevant for clinical trials in mechanical circulatory support. Finally, cardiac arrest presents a

unique challenge in the study of CS and is a subject of ongoing debate. Addressing the criteria for CS may be the first step towards resolving these debates.

Conclusion

Cardiogenic shock is associated with a high risk of mortality, with little improvement in survival over the last few decades. A review of the criteria for the diagnosis of CS is timely, as clinicians and trialists have adopted different criteria for CS, complicating interpretation of clinical studies and clinical application. These considerations of this European group of experts will provide greater uniformity in the diagnosis of CS and strongly support a *Universal Definition of Cardiogenic Shock*.⁷⁸

Author contributions

Hoong Sern Lim (Conceptualization, Writing—original draft [equal]), Agnieszka Tycinska (Writing—review & editing [equal]), Clement Delmas (Writing—review & editing [equal]), Jacob Eifer Moller (Writing—review & editing [equal]), Janine Pöss (Writing—review & editing [equal]), Christian Stoppe (Writing—review & editing [equal]), Christophe Vandenbriele (Writing—review & editing [equal]), Holger Thiele (Writing—review & editing [equal]), Kurt Huber (Writing—review & editing [equal]), Ingo Ahrens (Writing—review & editing [equal]), Christian Hassager (Writing—review & editing [equal]), Ovidiu Chioncel (Writing—review & editing [equal]), Marco Metra (Writing—review & editing [equal]), Matteo Pagnesi (Writing—review & editing [equal]), Wiktor Kuliczowski (Writing—review & editing [equal]), Maja Cikes (Writing—review & editing [equal]), Linda W. Van Laake (Writing—review & editing [equal]), Alessandro Sionis (Writing—review & editing [equal]), Mario Iannaccone (Writing—review & editing [equal]), Stephane Manzo-Silberman (Writing—review & editing [equal]), Federico Pappalardo (Writing—review & editing [equal]), Susanna Price (Writing—review & editing [equal]), Thomas F Lüscher (Writing—review & editing [equal]), and Benedikt Schrage (Conceptualization, Writing—original draft [equal], Writing—review & editing [lead])

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